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GOOGLE PROMPT ENGINEERING COURSE



PROMPT COMPARISON ENGINE

NAZSYA AI TUTOR

AI Smart Tutor Using Advanced Prompt Engineering Techniques

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A. Prompt Comparison Engine and Optimization

1. Prompt Comparison Engine Development

In this phase, the project developed a Prompt Comparison Engine to compare the quality of AI-generated educational responses using different prompt engineering techniques.

The initial prototype tested several prompting strategies, including Standard Prompting, Basic Prompting, Structured Prompting, Few-Shot Prompting, and Optimized Prompting. This comparison was used to analyze differences in response quality, structure, consistency, and readability.

In the final OpenRouter-based implementation, the comparison engine focused on comparing the Basic Prompt Response and the Optimized Prompt Response. This approach allowed the project to clearly evaluate how optimized prompt engineering improved the educational quality of the AI tutor response.

1.1 Initial Multiple Prompt Generation Implementation

The following code shows the initial prompt comparison implementation used during the early prototype stage.

```
# Generate Responses

standard_response = model.generate_content(prompt)

basic_response = model.generate_content(basic_prompt)

structured_response = model.generate_content(structured_prompt)

few_shot_response = model.generate_content(few_shot_prompt)

optimized_response = model.generate_content(optimized_prompt)
```

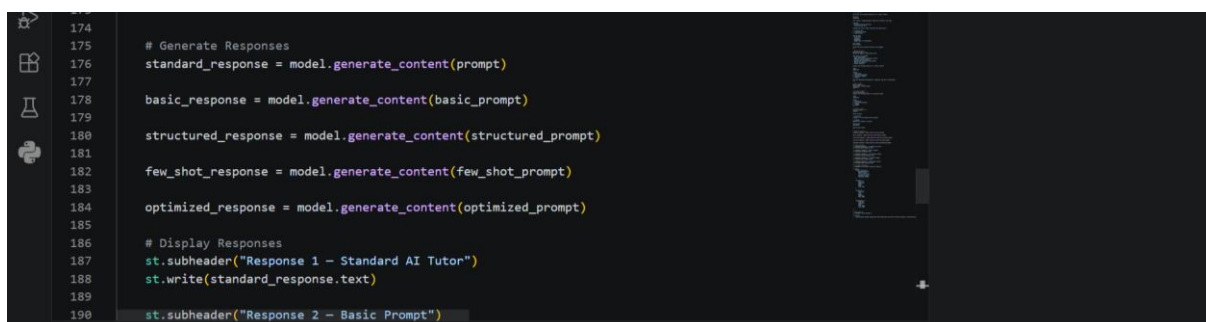


Figure 1.1 Multiple prompt generation implementation using several prompt engineering techniques

1.2 Prompt Comparison Display Implementation

```
st.subheader("Response 1 - Standard AI Tutor")
st.write(standard_response.text)

st.subheader("Response 2 - Basic Prompt")
st.write(basic_response.text)

st.subheader("Response 3 - Structured Prompt")
```

```

st.write(structured_response.text)

st.subheader("Response 4 – Few-Shot Prompt")
st.write(few_shot_response.text)

st.subheader("Response 5 – Optimized Prompt")
st.write(optimized_response.text)

```

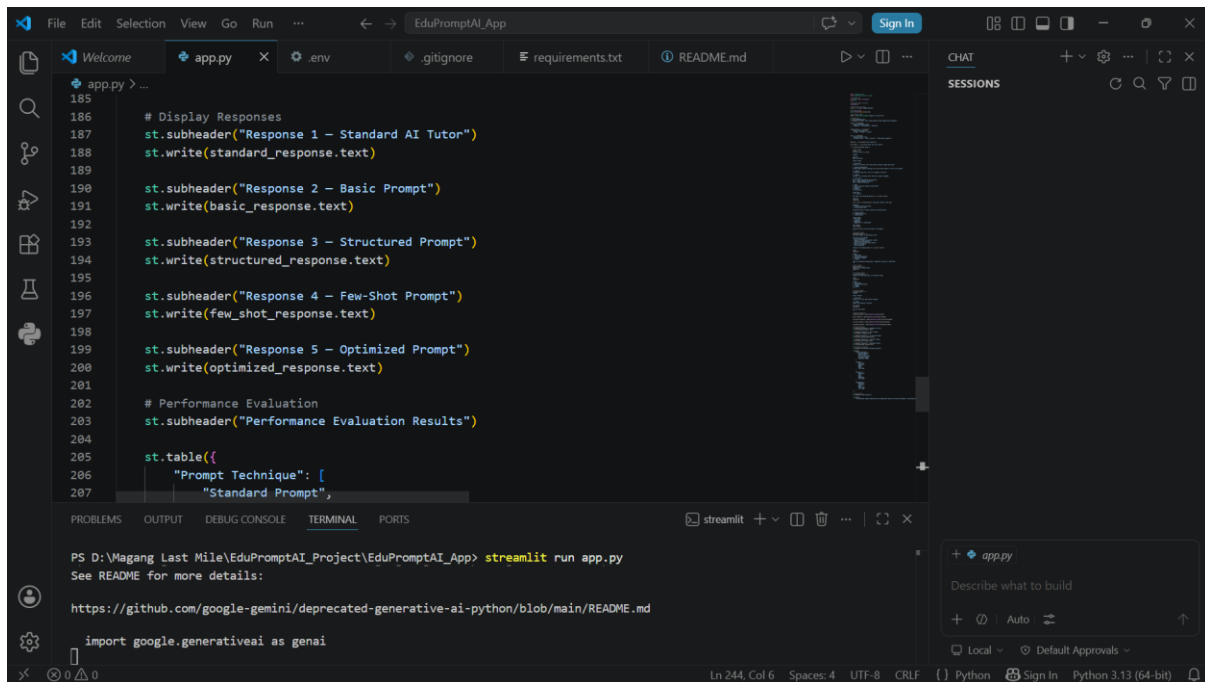


Figure 1.2 Comparison of multiple AI-generated responses using different prompt engineering techniques

1.3 Final Prompt Comparison Engine Implementation in OpenRouter Version

After the initial prompt comparison engine was tested, the feature was adapted into the final version of Nazsya AI Tutor using OpenRouter API and the DeepSeek Chat model.

In the final implementation, the system compares two main response types: Basic Prompt Response and Optimized Prompt Response. The Basic Prompt Response generates a short and general explanation, while the Optimized Prompt Response applies multiple prompt engineering techniques, including structured prompting, learner-level adaptation, few-shot prompting, study plan generation, quiz generation, and role-based instructional guidance.

This final comparison helps demonstrate the improvement produced by prompt optimization in terms of structure, completeness, personalization, and educational usefulness.

```

basic_prompt = f"""
Explain this topic briefly:

{question}
"""

basic_response = client.chat.completions.create(
    model="deepseek/deepseek-chat",
    messages=[
        {
            "role": "user",
            "content": basic_prompt
        }
    ]
)

basic_output = basic_response.choices[0].message.content

response = client.chat.completions.create(
    model="deepseek/deepseek-chat",
    messages=[
        {
            "role": "user",
            "content": optimized_prompt
        }
    ]
)

optimized_response = response.choices[0].message.content

```

Figure 1.3 Basic prompt response implementation using OpenRouter API

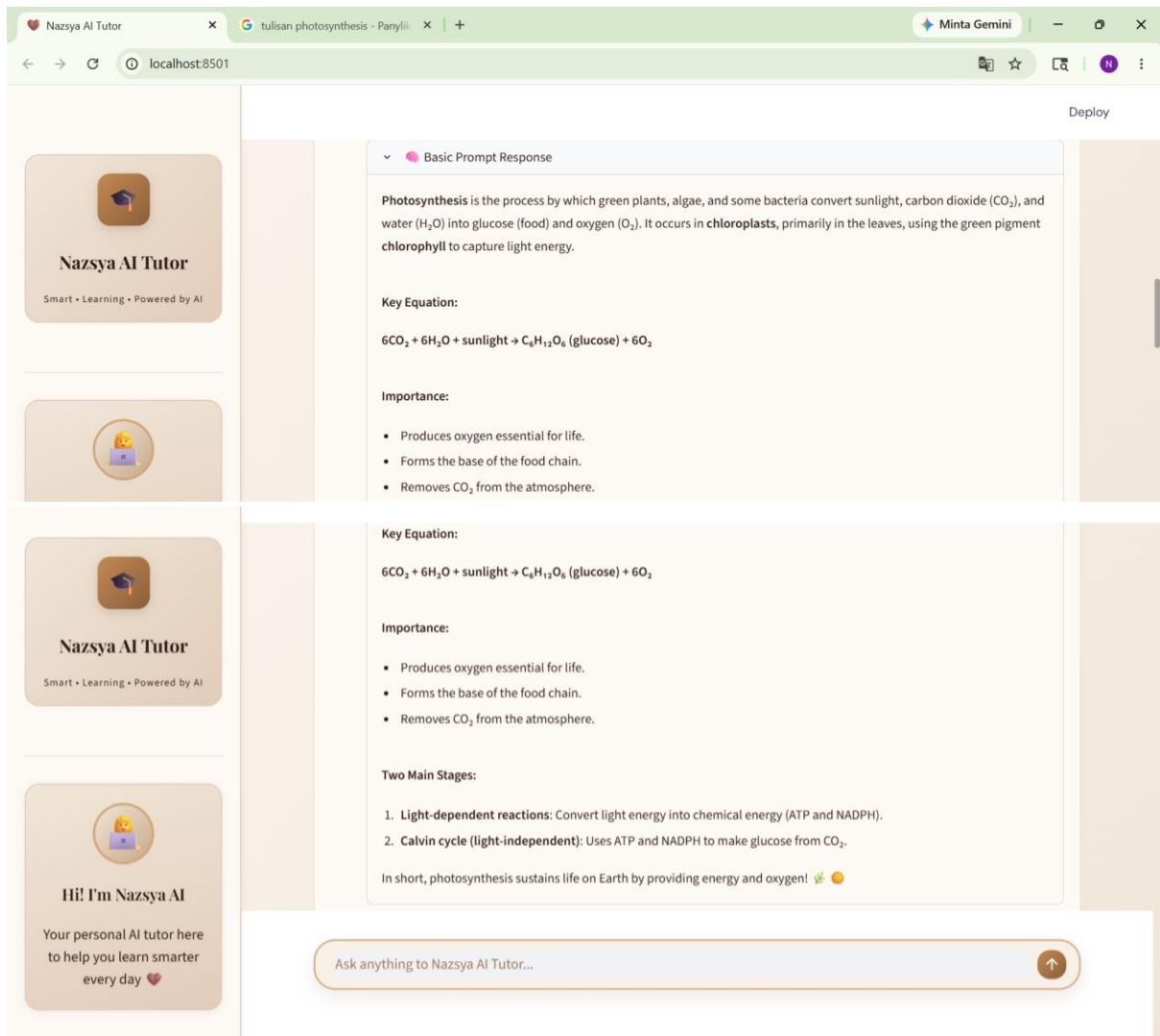
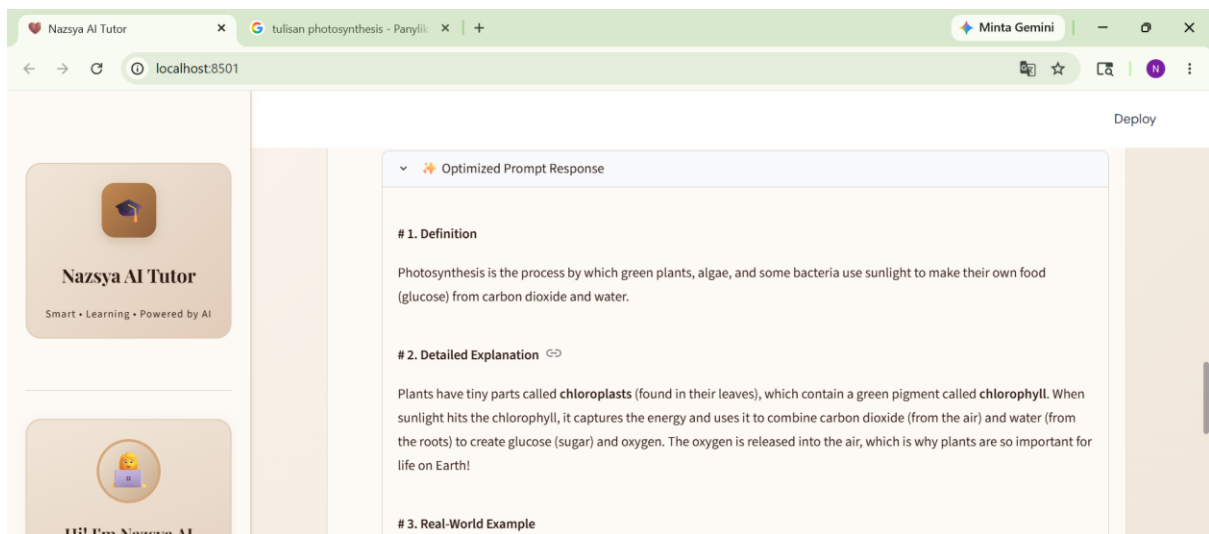


Figure 1.4 *Optimized prompt response implementation using OpenRouter API*



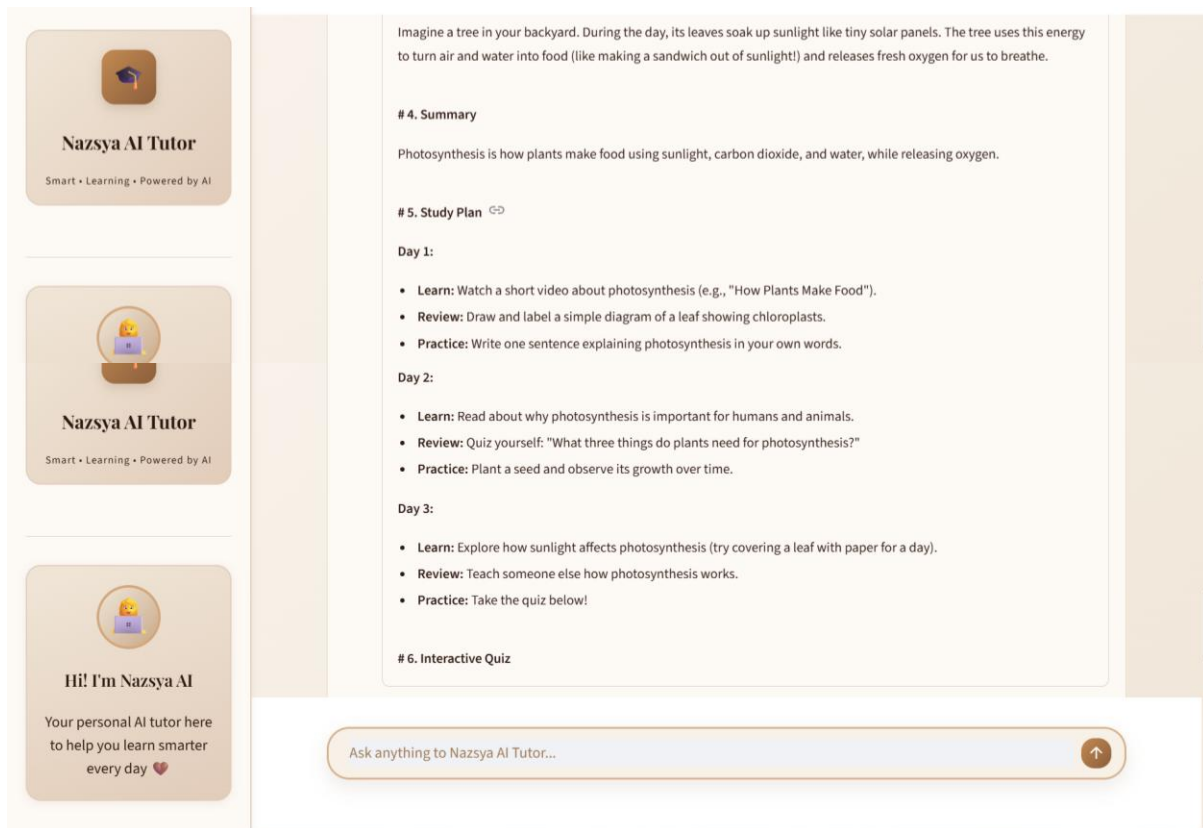


Figure 1.5 Final comparison display between Basic Prompt Response and Optimized Prompt Response

Summary of prompt comparison output:

The Basic Prompt Response generated a general explanation of photosynthesis. It explained the definition, equation, importance, and main stages of photosynthesis. Although the response was useful, it was still presented as a general explanation and did not provide a complete learning structure.

The Optimized Prompt Response generated a more organized educational response. It included structured sections such as Definition, Detailed Explanation, Real-World Example, Summary, and Study Plan. The explanation was also more learner-friendly and used simple analogies to make the topic easier to understand.

This comparison shows that optimized prompting improved the structure, readability, and educational usefulness of the AI-generated response compared to the basic prompt.

1.4 Prompt Comparison Analysis

The prompt comparison experiment demonstrated that different prompt engineering strategies significantly affected the quality of AI-generated educational responses.

The Basic Prompt Response produced a short and general explanation of the topic. It was useful for quick understanding, but it did not include structured sections, learner-level adaptation, study plans, quizzes, or interactive evaluation.

In contrast, the Optimized Prompt Response generated a more complete educational output. It included Definition, Detailed Explanation, Real-World Example, Summary, Study Plan, and

Interactive Quiz. The optimized response was also adapted based on the selected learner level and evaluator role.

This shows that optimized prompting improved response structure, completeness, personalization, and educational usefulness compared to basic prompting.

The following table summarizes the comparison between the Basic Prompt Response and the Optimized Prompt Response in the final application.

Aspect	Basic Prompt Response	Optimized Prompt Response
Response Style	Short and general	Structured and educational
Output Format	Paragraph-based explanation	Definition, Explanation, Example, Summary, Study Plan, and Quiz
Learner Level Adaptation	Not available	Available
Study Plan	Not available	Available
Quiz	Not available	Available
Educational Usefulness	Basic understanding	Complete learning support
Prompt Engineering Technique	Basic Prompting	Optimized Prompting with multiple techniques

Based on the prompt comparison engine testing, the project concluded that optimized prompting produced more structured, complete, and educationally useful responses than basic prompting. The comparison engine helped demonstrate the impact of advanced prompt engineering techniques in improving the quality of AI-generated learning content.

2. Prompt Optimization

In this phase, the project optimized the prompt structure to improve educational response quality, clarity, organization, consistency, and personalization.

The optimization process was carried out by adding clearer instructions, structured response sections, learner-level adaptation, few-shot prompting examples, study plan generation, quiz generation, and role-based behavior rules.

In the final OpenRouter-based implementation, the optimized prompt guided the DeepSeek Chat model to generate more complete and educational responses. The optimized response was designed not only to explain a topic, but also to support learning through examples, summaries, study plans, quizzes, and role-based instructional guidance.

2.1 Initial Optimized Prompt Implementation

The following code shows the initial optimized prompt design used to improve the structure and clarity of AI-generated educational responses.

```
optimized_prompt = f"""
You are an expert AI educational tutor.
```

```

Your task is to provide:
- Clear explanations
- Beginner-friendly educational content
- Step-by-step explanations
- Concise and well-structured answers
- Real-world examples
- Simple summaries

Explain the following topic for a {level} learner:

Topic:
{question}

Format:
1. Definition
2. Detailed Explanation
3. Real-World Example
4. Summary

Keep the explanation educational, organized, and easy to understand.
"""

```

```

111 # Optimized Prompt
112 optimized_prompt = f"""
113 You are an expert AI educational tutor.
114
115 Your task is to provide:
116 - Clear explanations
117 - Beginner-friendly educational content
118 - Step-by-step explanations
119 - Concise and well-structured answers
120 - Real-world examples
121 - Simple summaries
122
123 Explain the following topic for a {level} learner:
124
125 Topic:
126 {question}
127
128 Format:
129 1. Definition
130 2. Detailed Explanation
131 3. Real-World Example
132 4. Summary

```

Figure 2.1 *Optimized prompt implementation for improving educational response quality*

Response 5 — Optimized Prompt

Hello there! I'm your AI educational tutor, and I'm excited to help you understand what photosynthesis is. Don't worry, we'll break it down step-by-step!

1. Definition

Photosynthesis is the process that plants, algae, and some bacteria use to convert light energy into chemical energy in the form of food (sugars). Think of it as how plants make their own snacks using sunlight!

2. Detailed Explanation

Let's break down how photosynthesis works:

- **The Ingredients:** Plants need three main things to perform photosynthesis:
 - **Sunlight:** This is the energy source. Plants have a special green pigment called **chlorophyll** (found mostly in their leaves) that's really good at capturing sunlight.
 - **Water (H₂O):** Plants absorb water from the soil through their roots.
 - **Carbon Dioxide (CO₂):** Plants take in carbon dioxide from the air through tiny pores on their leaves called stomata.

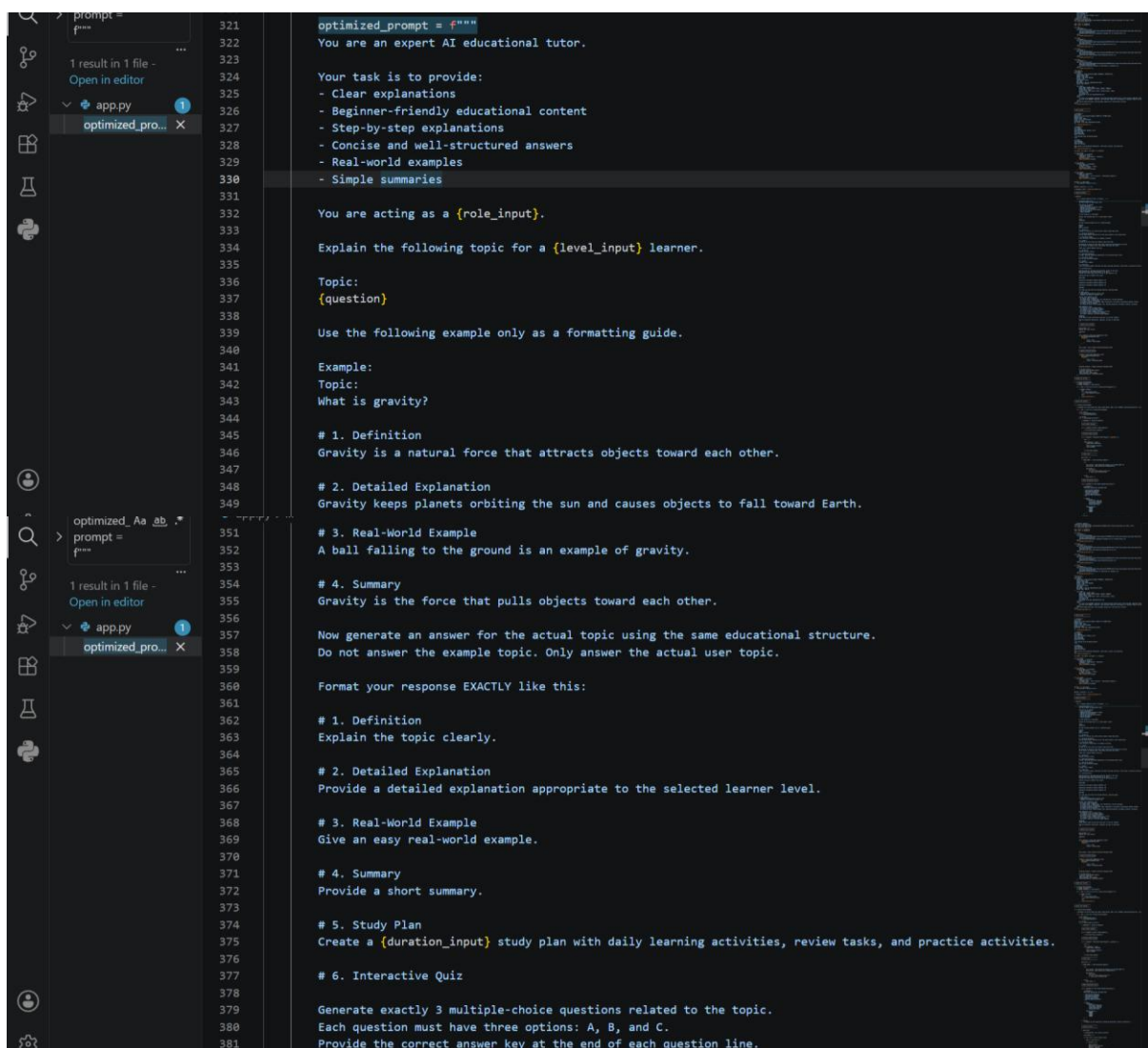
Figure 2.2 *AI-generated educational response using optimized prompt engineering techniques*

2.2 Final Optimized Prompt Implementation in OpenRouter Version

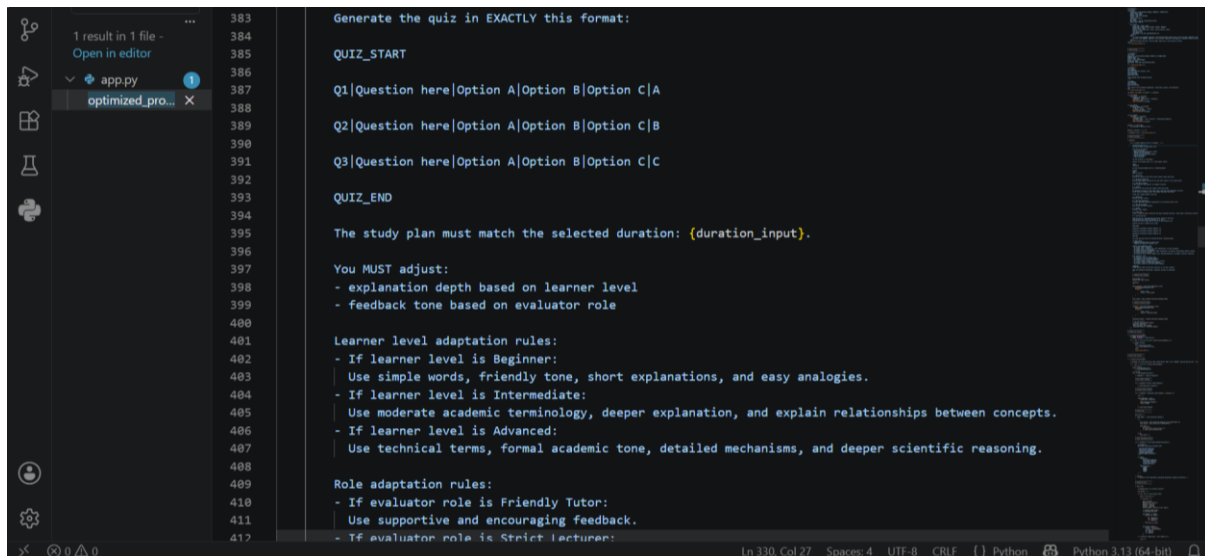
After the initial optimized prompt was tested, the prompt structure was further improved and implemented in the final version of Nazsya AI Tutor using OpenRouter API and the DeepSeek Chat model.

The final optimized prompt combined several prompt engineering techniques, including structured prompting, few-shot prompting, learner-level adaptation, study plan generation, interactive quiz generation, and role-based instruction.

This final prompt allowed the system to generate a more complete educational response. The output included Definition, Detailed Explanation, Real-World Example, Summary, Study Plan, and Interactive Quiz. The response style was also adjusted based on the selected learner level and evaluator role.



```
321 optimized_prompt = """
322 You are an expert AI educational tutor.
323
324 Your task is to provide:
325 - Clear explanations
326 - Beginner-friendly educational content
327 - Step-by-step explanations
328 - Concise and well-structured answers
329 - Real-world examples
330 - Simple summaries
331
332 You are acting as a {role_input}.
333
334 Explain the following topic for a {level_input} learner.
335
336 Topic:
337 {question}
338
339 Use the following example only as a formatting guide.
340
341 Example:
342 Topic:
343 What is gravity?
344
345 # 1. Definition
346 Gravity is a natural force that attracts objects toward each other.
347
348 # 2. Detailed Explanation
349 Gravity keeps planets orbiting the sun and causes objects to fall toward Earth.
350
351 # 3. Real-World Example
352 A ball falling to the ground is an example of gravity.
353
354 # 4. Summary
355 Gravity is the force that pulls objects toward each other.
356
357 Now generate an answer for the actual topic using the same educational structure.
358 Do not answer the example topic. Only answer the actual user topic.
359
360 Format your response EXACTLY like this:
361
362 # 1. Definition
363 Explain the topic clearly.
364
365 # 2. Detailed Explanation
366 Provide a detailed explanation appropriate to the selected learner level.
367
368 # 3. Real-World Example
369 Give an easy real-world example.
370
371 # 4. Summary
372 Provide a short summary.
373
374 # 5. Study Plan
375 Create a {duration_input} study plan with daily learning activities, review tasks, and practice activities.
376
377 # 6. Interactive Quiz
378
379 Generate exactly 3 multiple-choice questions related to the topic.
380 Each question must have three options: A, B, and C.
381 Provide the correct answer key at the end of each question line.
```



```
383 Generate the quiz in EXACTLY this format:
384
385 QUIZ_START
386
387 Q1|Question here|Option A|Option B|Option C|A
388
389 Q2|Question here|Option A|Option B|Option C|B
390
391 Q3|Question here|Option A|Option B|Option C|C
392
393 QUIZ_END
394
395 The study plan must match the selected duration: {duration_input}.
396
397 You MUST adjust:
398 - explanation depth based on learner level
399 - feedback tone based on evaluator role
400
401 Learner level adaptation rules:
402 - If learner level is Beginner:
403   Use simple words, friendly tone, short explanations, and easy analogies.
404 - If learner level is Intermediate:
405   Use moderate academic terminology, deeper explanation, and explain relationships between concepts.
406 - If learner level is Advanced:
407   Use technical terms, formal academic tone, detailed mechanisms, and deeper scientific reasoning.
408
409 Role adaptation rules:
410 - If evaluator role is Friendly Tutor:
411   Use supportive and encouraging feedback.
412 - If evaluator role is Strict Lecturer:
```

Figure 2.3 Final optimized prompt implementation using OpenRouter API

2.3 Prompt Optimization Analysis

The optimized prompt produced more educational, structured, and personalized AI responses compared to the basic prompt.

The final optimized prompt improved response quality by giving the AI clear instructions about output format, learner level, study duration, quiz format, and response style. As a result, the generated response became more organized and useful for learning.

Compared to the basic prompt, the optimized prompt produced a more complete educational output because it included structured explanations, real-world examples, summaries, study plans, and interactive quiz content.

Aspect	Basic Prompt	Final Optimized Prompt
Structure	General explanation	Highly structured educational format
Educational Clarity	Moderate	High
Readability	Good	Very good
Example Quality	Basic	Real-world contextual example
Learner Adaptation	Not available	Available
Study Plan	Not available	Available
Quiz Generation	Not available	Available
Few-Shot Guidance	Not available	Available
Consistency	Moderate	High

Based on the prompt optimization testing, the project concluded that the final optimized prompt improved the quality of AI-generated educational responses. The optimized prompt produced more structured, complete, consistent, and learner-friendly outputs compared to the basic prompt. This shows that prompt optimization plays an important role in improving the performance of Nazsya AI Tutor as an AI-powered educational assistant.

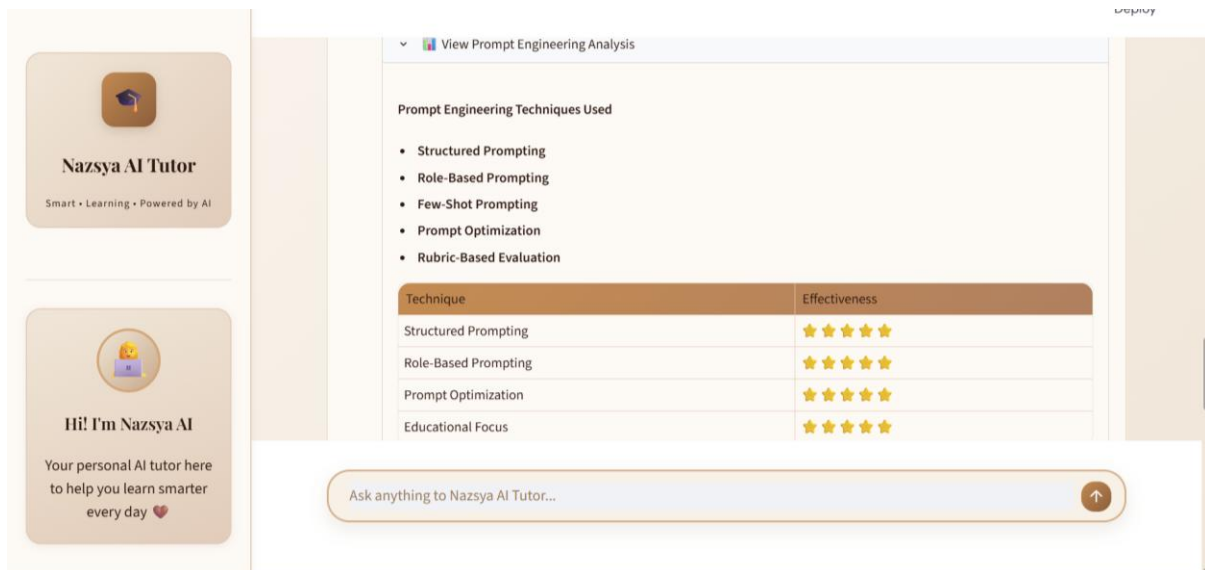


Figure 2.4 Prompt engineering analysis of techniques used in the final Nazsya AI Tutor application

The prompt engineering analysis section shows the main techniques implemented in the final Nazsya AI Tutor application, including Structured Prompting, Role-Based Prompting, Few-Shot Prompting, Prompt Optimization, and Rubric-Based Evaluation. This section helps summarize the prompt engineering strategies used to improve the quality of AI-generated educational responses.

3. Performance Evaluation

In this phase, the project evaluated the effectiveness of multiple prompt engineering techniques implemented in Nazsya AI Tutor. The evaluation focused on educational response quality metrics, including clarity, structure, and consistency.

The purpose of this evaluation was to compare how different prompting strategies affected the quality of AI-generated educational responses. The evaluated prompt techniques included Standard Prompting, Basic Prompting, Structured Prompting, Few-Shot Prompting, and Optimized Prompting.

The evaluation results showed that Optimized Prompting generated the highest overall educational response quality because it combined multiple prompt engineering techniques into one complete educational response format.

3.1 Performance Evaluation Implementation

The following code shows the performance evaluation implementation used to compare the effectiveness of several prompt engineering techniques.

The evaluation table compares five prompt techniques: Standard Prompt, Basic Prompt, Structured Prompt, Few-Shot Prompt, and Optimized Prompt. Each technique was evaluated based on three criteria: Clarity, Structure, and Consistency.

```
st.subheader("Performance Evaluation Results")
```

```

st.table({
    "Prompt Technique": [
        "Standard Prompt",
        "Basic Prompt",
        "Structured Prompt",
        "Few-Shot Prompt",
        "Optimized Prompt"
    ],

    "Clarity": [
        "Moderate",
        "Moderate",
        "High",
        "High",
        "Very High"
    ],

    "Structure": [
        "Moderate",
        "Low",
        "High",
        "Moderate",
        "Very High"
    ],

    "Consistency": [
        "Moderate",
        "Moderate",
        "High",
        "Very High",
        "Very High"
    ]
})

```

```

19
20 # Streamlit UI
21 st.title("EduPrompt AI")
22 st.subheader("AI Smart Tutor using Advanced Prompt Engineering Techniques")
23
24 level = st.selectbox(
25     "Choose explanation level",
26     ["Beginner", "Intermediate", "Advanced"]
27 )
28
29 study_duration = st.selectbox(
30     "Select study duration",
31     ["3 Days", "5 Days", "7 Days"]
32 )
33
34 role = st.selectbox(
35     "Choose evaluator role",
36     ["Friendly Tutor", "Strict Lecturer", "Professional Examiner"]
37 )
38
39 question = st.text_input("Ask a question")
40
41 user_prompt = st.text_area("Enter your prompt here")

```

PS D:\Magang Last Mile\EduPromptAI_Project\EduPromptAI_App> streamlit run app.py

See README for more details:

<https://github.com/google-gemini/deprecated-generative-ai-python/blob/main/README.md>

```

import google.generativeai as genai

```

Figure 3.1 Performance evaluation implementation for prompt engineering comparison

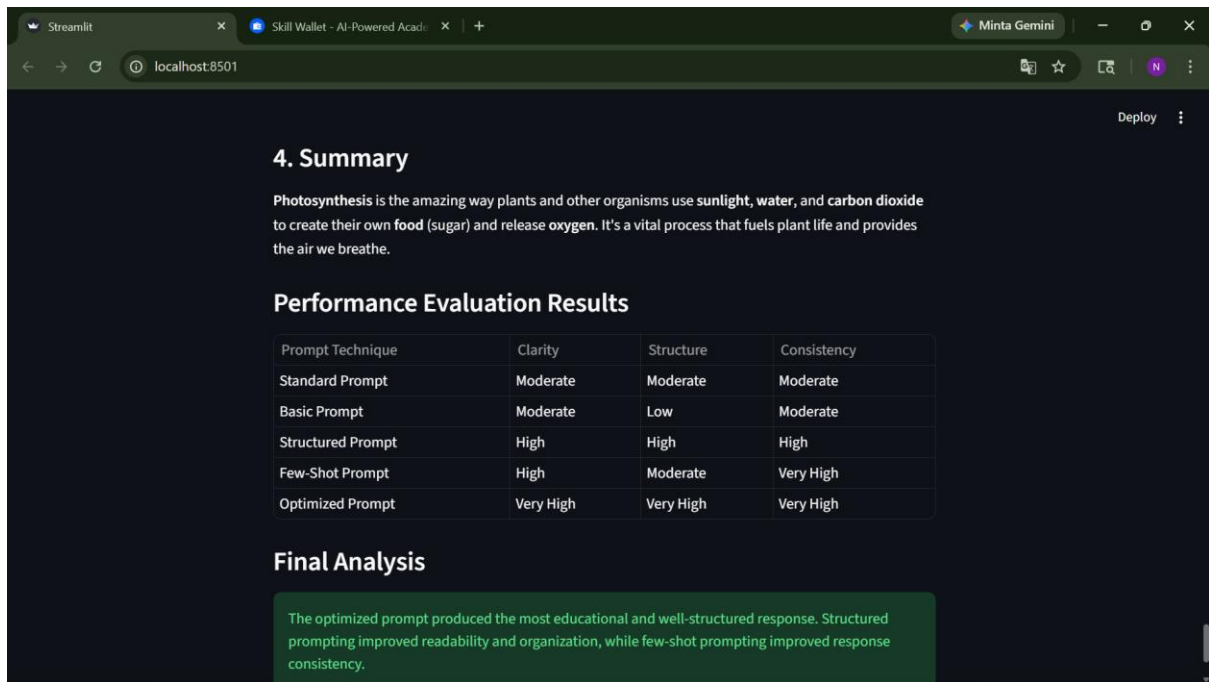


Figure 3.2 Performance evaluation results and final analysis of multiple prompt engineering techniques

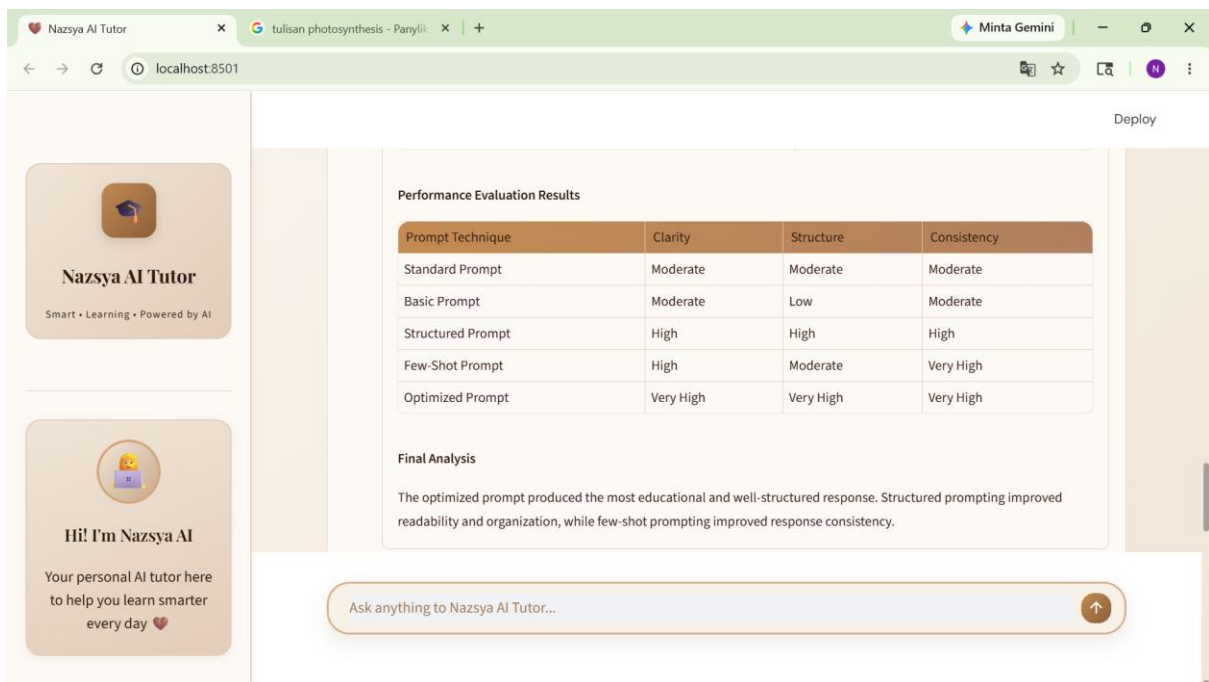


Figure 3.3 Performance evaluation results of multiple prompt engineering techniques

Summary of performance evaluation output:

The performance evaluation compared several prompt engineering techniques, including Standard Prompting, Basic Prompting, Structured Prompting, Few-Shot Prompting, and Optimized Prompting.

The Standard Prompt generated a general educational response, while the Basic Prompt produced a longer but less structured explanation. The Structured Prompt improved organization by separating the response into clearer sections. The Few-Shot Prompt produced a more consistent format by following the provided example.

The Optimized Prompt produced the strongest result because it combined structured explanation, learner-friendly language, real-world examples, summary, study plan, and quiz generation. Based on the evaluation table, the Optimized Prompt achieved Very High ratings in clarity, structure, and consistency.

3.2 Performance Evaluation Analysis

The performance evaluation results confirmed that prompt engineering significantly influenced AI-generated educational responses.

Standard Prompting produced a general response, while Basic Prompting generated a longer explanation but with weaker structure. Structured Prompting improved response organization and readability. Few-Shot Prompting improved consistency because the AI followed the provided example format.

Optimized Prompting produced the best overall result because it combined several techniques, including structured prompting, few-shot prompting, learner-level adaptation, study plan generation, quiz generation, and role-based evaluation.

Based on the performance evaluation table, the Optimized Prompt achieved Very High ratings in clarity, structure, and consistency. This shows that prompt optimization improved the instructional effectiveness and educational quality of Nazsya AI Tutor.

The following table summarizes the best-performing prompt technique based on each evaluation aspect.

Evaluation Aspect	Best Performing Prompt	Reason
Clarity	Optimized Prompt	Provides clear and learner-friendly explanation
Structure	Optimized Prompt	Uses organized sections such as Definition, Explanation, Example, Summary, Study Plan, and Quiz
Consistency	Optimized Prompt	Follows a stable prompt format and output structure
Educational Usefulness	Optimized Prompt	Supports explanation, practice, study planning, and evaluation
Personalization	Optimized Prompt	Adapts response based on learner level and evaluator role

Based on the performance evaluation, the project concluded that optimized prompting produced the highest educational response quality compared to other prompt engineering

techniques. The optimized prompt improved clarity, structure, consistency, and overall learning usefulness.

This evaluation demonstrated that advanced prompt engineering techniques play an important role in improving the performance of Nazsya AI Tutor as an AI-powered educational assistant.

Overall, the Prompt Comparison Engine successfully demonstrated the difference between basic prompting and optimized prompting in Nazsya AI Tutor. The comparison results showed that optimized prompting produced clearer, more structured, more consistent, and more educationally useful responses. Therefore, this optional component supports the main objective of the project by proving that advanced prompt engineering techniques can improve AI-generated learning content.